

IS508 - Big Data and Analytics

Group Assignment

Instructions:

Please read and solve the attached case.

Each group please prepare:

1. A **presentation**, 5 – 7 minutes long, with your analysis and recommendation. Take on the role of Sally Bunn.
2. By the deadline, turn in your presentation with summary of your analysis, answers to the questions in the case and including the Excel output of your best prediction model as the last page of your presentation.
No separate Excel file will be accepted.

This project counts 20% of your total grade.

Assessment Criteria:

The main assessment criteria will be:

- Quality of the presentation
- Development and Analysis of the model
- Answers to the case questions

Magic Kitchens Meat Loaf Mix

United Food Products (UFP) markets a meat loaf mix under the brand name “Magic Kitchens” in far western United States. The mix is packaged in foil packets, and contains cracker crumbs, meal, seasonings and other ingredients. The purchaser mixes the ingredients together with an egg and ground beef to make a meat loaf. The mix has the effect of stretching the meat loaf, as well as seasoning it.

The Magic Kitchen meat loaf mix is an established brand, and although its sales are not large, it is a consistently profitable item. Sally Bunn has just joined UFP as brand manager, and the meat loaf product was assigned to her. Her first task is to prepare a sales forecast, and a budget for promotion and advertising for the next year. She collected the historical data shown in the table on the next page. These data included sales of the meat loaf mix, and also the expenditures for promotion and advertising over the past 24 quarters, all in thousands of dollars. Also included is an index of “general economic conditions” in the meat loaf market area. High values of the index indicate “good” economic times.

The meat loaf mix is sold through food brokers in Seattle, San Francisco, Los Angeles, and Denver. Advertising expenditures are directed at the consumer in magazines such as *Women’s Day*, and in newspapers. Promotion expenditures, on the other hand, are directed at the food broker or store manager. These consist of special deals such as getting a fifth case free if four were purchased, short term increases in broker commissions, or sales contests among broker salesmen (with prizes such as trips to Hawaii).

Ms. Bunn was quite puzzled by the great variability in the sales of the meat loaf mix from quarter to quarter, and also at the great variations in past expenditures for promotion and advertising. Upon inquiry, the sales vice president explained that there was a general policy that the company should either promote or advertise in a given quarter, but not both. However, there had been a long-standing dispute in the company about the relative effectiveness of promotion and advertising on meat loaf mix sales. Her predecessors had tried various different things, but no one had been able to determine what had and what had not been successful. Some sceptics felt that both promotion and advertising expenditure were wasted, since neither seemed to affect sales. Others felt that promotion was effective, but the effect was merely to reduce future sales. That is, they felt that brokers and store managers bought heavily during promotion periods, and then didn’t order in subsequent periods until inventories were back to normal. The effect of advertising was equally confusing, since sales seemed to vary greatly even during periods when advertising was relatively constant. For example, in the last two quarters in the data set shown below (quarters 23 and 24), advertising had been about the same (36 and 39 thousand respectively), but sales were \$648 thousand in one quarter and only \$343 thousand in the other.

If this wasn’t confusing enough, the economist in the Economic Analysis Department insisted that the meat loaf mix was a “counter-cyclical” product, meaning that it sold better in bad times than in good. His theory was that meat loaf was a less expensive meal than other meats, and people ate more of it during tough times. Furthermore, he felt that there was a seasonal pattern in sales, with more being sold during colder months than in the summer.

Basic Data for Magic Kitchens Meat Loaf Mix (Mkitch.xls)

obs	sales	prom	adv	index
1	504.72	15.6	30	100
2	406.59	22.2	36	102
3	398.55	0.0	45	104
4	587.76	0.0	57	104
5	598.92	0.0	39	104
6	703.62	31.8	21	100
7	387.24	21.3	12	98
8	365.67	3.9	6	96
9	388.71	0.0	6	98
10	372.96	8.4	30	103
11	603.30	45.3	30	105
12	614.73	50.1	33	107
13	484.38	39.6	6	107
14	227.76	4.2	33	107
15	329.13	0.0	6	108
16	308.25	0.0	3	105
17	433.86	0.0	45	103
18	514.98	13.8	48	108
19	404.70	17.7	0	110
20	245.43	0.0	15	112
21	433.20	17.4	9	113
22	627.24	37.8	54	112
23	647.61	42.3	36	113
24	342.81	11.4	39	114
Mean	455.51	16.0	26.6	105.5

sales - quarterly sales of Magic Kitchens meat loaf mix (thousands of dollars)

prom - funds spent on promotion activities in the quarter (thousands of dollars)

adv - funds spent on advertising during the quarter (thousands of dollars)

index - economic index of general economic conditions in Magic Kitchens market area

The data is available in an Excel file Mkitch.xls on Moodle with variable names in the first row.

You may wish to consider that advertising and promotion could have a delayed effect on sales.

Questions

1. If Ms. Bunn had one thousand dollars to spend on either advertising or promotion, which should she choose, and why? What is the effect of one thousand dollars spent on each?
2. As described above, the economic analyst thought that the meat loaf mix was a “counter-cyclical” item. Would you agree?
3. Do you think there are significant seasonal effects associated with meat loaf mix sales? How do you know?